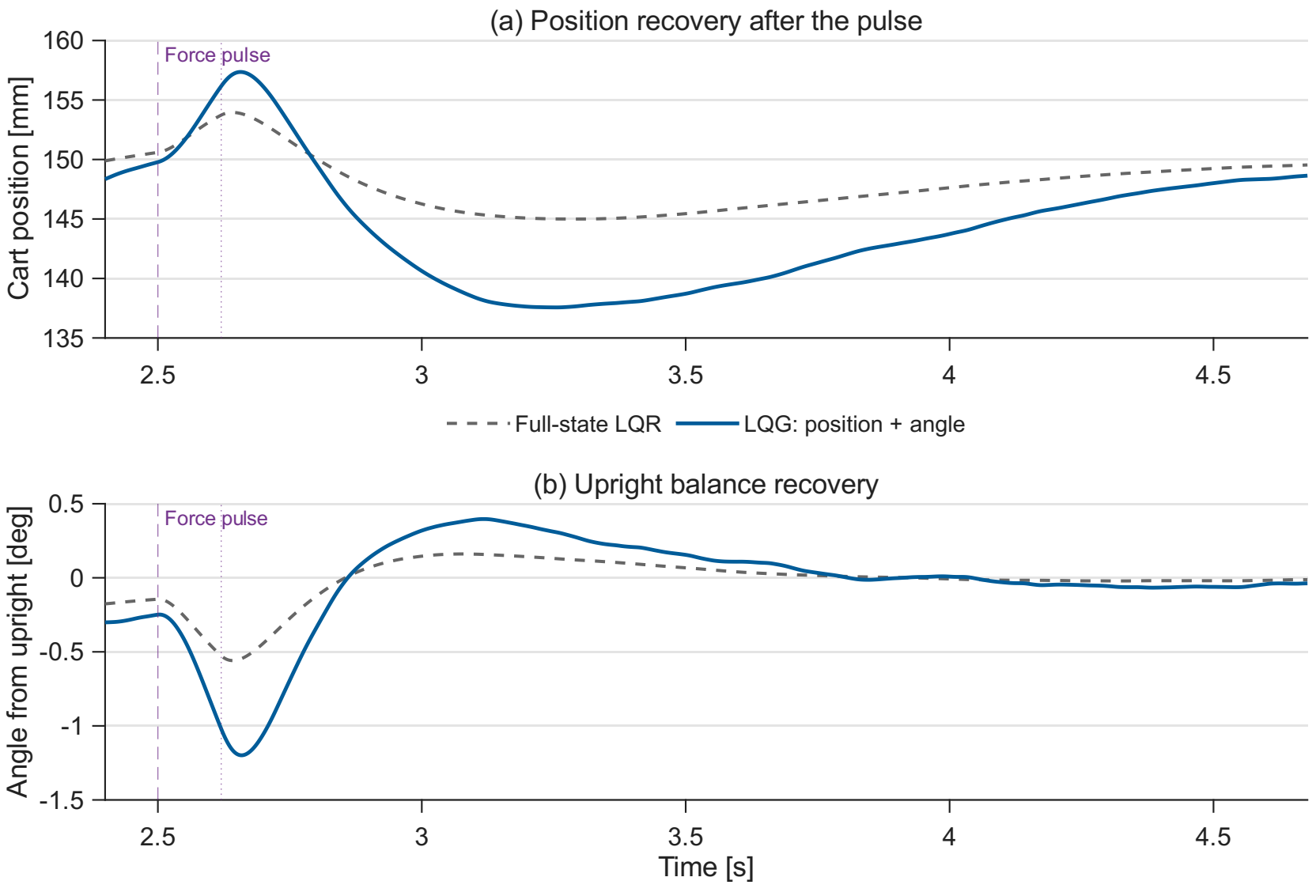

(a) Position recovery after the pulse
Force pulse
Cart position [mm]
160
155
150
145
140
135
2.5
3
3.5
4
4.5
Full-state LQR
LQG: position + angle
(b) Upright balance recovery
Force pulse
Angle from upright [deg]
0.5
0
-0.5
-1
-1.5
2.5
3
3.5
4
4.5
Time [s]